

Rahul Rawat

DATA PLATFORM AND AI ENGINEERING WORKSTUDENT

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PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on data platform and AI engineering work across cloud pipelines and LLM applications. I have built a PySpark and dbt driven real time flight tracking pipeline on Google Cloud, an automated BigQuery medallion architecture with a BigQuery ML classifier, and a Retrieval Augmented Generation system with agentic routing on Llama 3.1 8b via Groq. Comfortable in Python, SQL, PySpark, Git, and LLM tooling, and with a working Databricks style mindset, I am the right fit for driving data platform improvements and AI proofs of concept in parallel.

EXPERIENCE

eRay GmbH

Oct 2025 to Mar 2026

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

EDUCATION

M.Sc. Data Science and Analytics

Apr 2025 to Present

SRH University of Applied Sciences Heidelberg

Bachelor of Technology in Computer Science

2019 to 2023

GL Bajaj Institute of Technology and Management

PERSONAL PROJECTS

Real Time Flight Tracking Data Pipeline

Python PySpark BigQuery dbt Apache Airflow GCP Dataproc and GCS Tableau with TabPy

- Collected live flight positions from the OpenSky Network API every 30 seconds and enriched them against airport, aircraft, and weather data across four sources, measured by a clean joined table covering more than 128 thousand records, using Python collectors and PySpark cleaning on Google Cloud.
- Shaped raw collector output into analysis ready tables with dbt and computed each aircraft's nearest airport along the way, measured by consistent nearest airport labels across the historical dataset, using PySpark for heavy lift and dbt for the modelling layer.
- Orchestrated the whole system with Apache Airflow so batch and real time layers refresh automatically every 15 minutes, measured by unattended DAG runs on GCS backed storage and Dataproc compute.
- Surfaced findings in a Tableau workbook backed by Python statistics through TabPy, measured by the finding that air traffic drops 4.4 times in heavy rain and clusters around hubs like Frankfurt and Munich, using dbt aggregates as the visual layer feed.

Movie Analytics and ML Pipeline on GCP

GCP BigQuery

Cloud Run

GCS

Cloud Scheduler

BigQuery ML

Python

SQL

Looker Studio

- Built an end to end batch pipeline pulling movie data from a public API into a GCS data lake and processing it through Bronze to Silver to Gold medallion architecture in BigQuery, measured by a fully automated Cloud Scheduler trigger with no manual intervention required, using Cloud Run for compute.
- Engineered the Silver layer for trustworthy analytics with schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, measured by clean referential integrity across all downstream Gold tables.
- Trained a BigQuery ML classifier that predicts whether a film will be a hit before release, measured by leakage free evaluation, by deliberately splitting features into two tables so only pre release signals feed the model.
- Built Gold layer aggregates and a five page Looker Studio dashboard answering concrete business questions on genre ROI, foreign language growth, and release season timing, measured against the ML early warning view, and secured the system with a least privilege service account and Secret Manager.

Hybrid RAG Orchestrator with Agentic Routing

Python

LangChain

Llama 3.1 8b via Groq

ChromaDB

HuggingFace MiniLM L6 v2

Streamlit

- Delivered a modular Retrieval Augmented Generation system that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, measured by clean end to end responses across all three routes, using a custom decision making router on top of Llama 3.1 8b via Groq and LangChain orchestration.
- Built a persistent semantic memory over PDF and vector data, measured by consistent retrieval on cross session queries, using ChromaDB with local persistence and HuggingFace MiniLM L6 v2 embeddings.
- Kept multi turn context intact for the LLM, measured by coherent follow up answers rather than isolated single shot responses, by adding a stateful MemoryAgent directly into the inference pipeline.
- Shipped the whole system as a Streamlit interface with end to end ownership, measured by a working deployed prototype, using Groq inference and a modular routing layer that can be extended per new intent.

RESEARCH AND THESIS

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 patients, using 10 fold cross validation and per model confusion matrices to make the model comparison auditable.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.

CERTIFICATIONS

- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English: fluent, professional working proficiency. **German:** B1, in progress.